Udit Chhipa

Cyber Security & Vulnerability Research Enthusiast

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Education

Jaipur National University Jaipur, India
B.Tech, Computer Science (Cyber Security) • CGPA: 7.8 2023 – 2027

Experience

Rajasthan Police – Cyber Crime Branch Apr – May 2026
Cyber Security & AI Research Intern Jaipur

- Built LLM-powered agentic investigation systems and VLM-based weapon detection using domain-specific datasets.
- Designed ML models for cybercrime detection; Telegram OSINT and criminal network analysis pipelines.
- Presented research prototypes to Cyber Crime teams; received Letter of Recommendation from Rajasthan Police HQ.

ShadowFox (Cybersecurity Training & Services Firm) Jul – Aug 2025
Cyber Security Intern Remote

- Web app vulnerability assessments (Burp Suite, Nessus, Nmap); REST API testing for auth flaws and misconfigurations.
- OSINT reconnaissance for exposed assets; Wireshark-based network traffic analysis and incident reporting.

Security Research Projects

NEBULA – Threat Emulation Research [Rust, Windows API FFI] 2025–Present • (Private)

- Researched modern EDR/AV detection gaps via a threat-emulation implant exercising 11+ evasion techniques (AMSI/ETW patching, direct syscall stubs, module stomping, PPID spoofing, sleep/stack obfuscation) and 26+ sandbox/VM detection checks.
- Studied multi-channel C2 design (HTTPS/TCP/DNS/dead-drop via GitHub & Telegram/SMB) with auto-fallback; companion Android research implant in C.

BLACKOUT-WINDOWS – C2 Architecture Research [Python, C#, Windows Internals] 2024–Present • (Private)

- Researched Windows internals and detection evasion via a C2 research framework: web dashboard, agent builder, Hell's Gate syscall resolution, process hollowing, BYOVD discovery, UAC bypass research, and steganography-based payload concealment.

PHANTOM-INTEL – Threat Intelligence Platform [Python, OSINT, AI/ML] 2024–2025 • (Private)

- Automated dark web monitoring, BIN leak detection, MITRE ATT&CK attribution, geopolitical threat mapping across 190+ countries.

Technical Skills

- **Languages:** C (Basic) • C++ • Python
- **Offensive Security:** C2 Development • Implant Engineering • Process Injection • Syscall Research • UAC Bypass
- **Detection Evasion Research:** AV/EDR Evasion • AMSI/ETW Patching • Sleep Obfuscation • Stack Spoofing • Binary Obfuscation
- **Vuln Research:** Windows Internals • AFL++ Fuzzing • Ghidra Static Analysis • ACL Research
- **Threat Intel & OSINT:** MITRE ATT&CK • Dark Web Monitoring • Telegram OSINT • Intelligence Aggregation
- **AI Security:** LLM Workflows • VLM Systems • ML Malware Detection Research • Agentic AI
- **Problem Solving:** Data Structures & Algorithms • Analytical Thinking • Root-Cause Analysis
- **Tools:** Burp Suite • Wireshark • Nmap • Nessus • Metasploit • Ghidra • Docker

Achievements

- Letter of Recommendation from Rajasthan Police HQ for AI-assisted cybercrime investigation research.